

Unit 4 — Social Interactions

Game theory, social dilemmas & how we escape them

A companion study guide and running lecture notes. Every payoff matrix, game tree and graph from the slides is redrawn here so you can follow along visually while the lecturer talks. Definitions, intuition boxes, worked equilibrium solutions and exam-style tips are added throughout. Academic year 2025–2026.

HOW TO USE THESE NOTES

Read the **definition boxes** before the lecture to prime the vocabulary. During the lecture, follow the **redrawn diagrams** — they use the same colours and layout as the slides. After the lecture, test yourself with the **self-test questions** at the end. A recurring convention is used for every game: ■ **blue = the row player** (bottom-left of each cell) and ■ **magenta = the column player** (top-right of each cell).

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A. Introduction — why self-interest is not always enough

Everything in the earlier units modelled a person choosing *alone*: a consumer picking a basket, a firm picking output. Their best choice did not depend on what anyone else did. Unit 4 drops that assumption. Here **your best action depends on other people's actions**, and theirs depend on yours.

The invisible hand — and its limits

Adam Smith's famous claim is that self-interest, channelled through markets, can produce socially good outcomes without anyone *intending* it:

ADAM SMITH, 1776 — *THE WEALTH OF NATIONS*

"He generally, indeed, neither intends to promote the public interest, nor knows how much he is promoting it. [...] he intends only his own gain, and he is in this, as in many other cases, led by an **invisible hand** to promote an end which was no part of his intention."

Self-interest often *does* drive beneficial outcomes — entrepreneurship, innovation, efficient markets. But it can also be **harmful**. The two questions that organise this whole unit are:

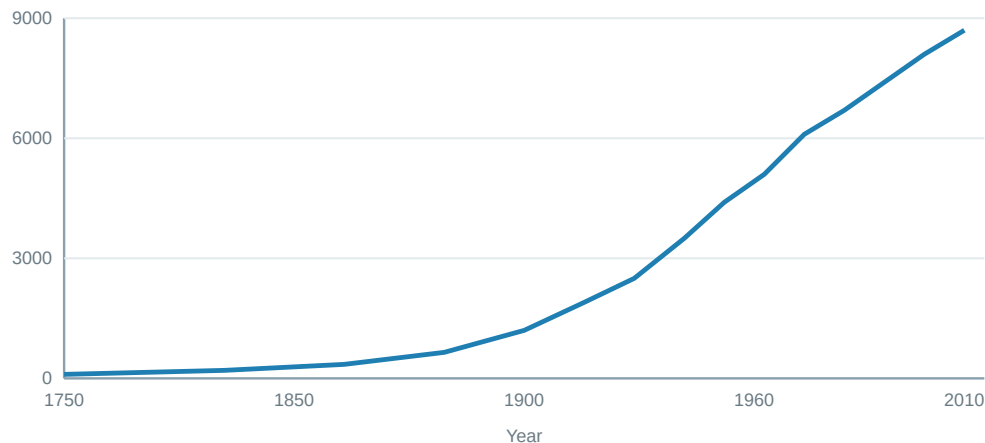
- **Why do these problems arise?** (What is it about the incentives?)
- **What can we do about it?** (Which mechanisms restore the good outcome?)

Social dilemmas

DEFINITION — SOCIAL DILEMMA

A situation in which actions taken **independently** by self-interested individuals result in a **socially sub-optimal outcome** — everyone (or society as a whole) could have been better off. Everyday examples: traffic jams, climate change, over-fishing, antibiotic resistance, the COVID-19 pandemic.

Carbon emissions from fossil fuels (a classic social dilemma)



Each driver, factory or country pursues its own interest; the cumulative effect is an outcome nobody chose. The steep post-1900 rise is the visual signature of a large-scale social dilemma.

Two archetypes and two cures

TRAGEDY OF THE COMMONS

Common property / shared resources tend to be **over-exploited** because each user ignores the cost imposed on everyone else. *W. F. Lloyd (1833)* used common grazing land → overgrazing; *Garrett Hardin (1968, Science)* generalised it. Modern versions: fishing, climate, the shared office kitchen.

FREE RIDING

Benefiting from others' contributions to a cooperative project **without contributing yourself**. If everyone free-rides, the project collapses — even though all would prefer it to succeed.

The engine behind both is the same: **people do not fully account for the effects of their actions on others**. The rest of the unit asks how **altruism**, **repeated interaction** and **government / institutional policy** can resolve these dilemmas.

B. Game theory — the key concepts

THREE BUILDING-BLOCK DEFINITIONS

- **Social interaction** — more than one party, where one person's action affects *both* their own and others' outcomes.
- **Strategic interaction** — a social interaction where people are *aware* that their actions affect others (and act on that awareness).
- **Strategy** — an action (or plan of actions) a person can take in a social interaction.

These interactions are studied with **game theory**, a toolkit used across the social sciences and even in biology.

What defines a "game"?

A game is a compact description of a social interaction with five ingredients. Learn them — every exam question that starts "set up the game" wants these five:

Element	Meaning
1. Players	Who is involved in the interaction.
2. Feasible strategies	The actions each player can take.
3. Order of play	Do players move at the same time (simultaneous) or one after another (sequential)?
4. Information	What each player knows when choosing their action.
5. Payoffs	The outcome (utility/profit) for <i>every</i> possible combination of actions.

Reading a payoff matrix — Example: Crop choice

Two farmers, **Anil** and **Bala**, each decide which crop to specialise in: *Rice* or *Cassava*. They interact only once (a **one-shot game**) and neither knows what the other will pick (a **simultaneous game**). Payoffs depend on market prices and land quality.

		Bala	
		Rice	Cassava
Anil	Rice	4 / 4	6 / 3
	Cassava	6 / 6	5 / 2

■ Anil (row) payoff, bottom-left · ■ Bala (column) payoff, top-right · ■ Nash equilibrium cell

HOW TO READ ANY CELL

Pick a row (Anil's choice) and a column (Bala's choice). The cell where they meet holds two numbers. The **bottom-left, blue** number is the **row player's** payoff; the **top-right, magenta** number is the **column player's** payoff. Example: if Anil grows Cassava and Bala grows Rice, the outcome is (Anil = 6, Bala = 6).

Nash equilibrium & best response

DEFINITIONS

- **Best response** — the strategy that gives a player the highest payoff, given the other player's strategy.
- **Nash equilibrium** — a set of strategies (one per player) such that *each* player's strategy is a best response to the others'. Named after **John Nash** (1928–2015, Nobel 1994; the film *A Beautiful Mind*).

Key property: in a Nash equilibrium no player can gain by unilaterally deviating (changing only their own strategy).

WORKED SOLUTION — FINDING THE NASH EQUILIBRIUM OF CROP CHOICE

Check each player's best response to every choice of the other. The **underline / circle method**: mark the best-response payoff in each row and column; a cell marked for *both* players is a Nash equilibrium.

If the other plays...	...the player's best response is	Because
Bala plays Rice	Anil → Cassava	$6 > 4$
Bala plays Cassava	Anil → Rice	$6 > 5$
Anil plays Rice	Bala → Rice	$4 > 3$
Anil plays Cassava	Bala → Rice	$6 > 2$

The mutual best responses meet at (**Anil = Cassava, Bala = Rice**) → payoff (**6, 6**). Neither wants to switch alone. That is the Nash equilibrium.

Here the Nash equilibrium also happens to be **Pareto efficient** (defined in §C) — but *this need not be true in general*, as the prisoner's dilemma below will show. Also note: **a game can have more than one Nash equilibrium** (see Coordination, §E).

A second example — practise the method

Player 1 chooses a row (T, M, B); Player 2 chooses a column (L, C, R). Find all Nash equilibria.

		Player 2		
		L	C	R
Player 1	T	2 / 1	0 / 2	0 / 3
	M	1 / 1	1 / 1	1 / 0
	B	0 / 1	2 / 0	2 / 2

■ Player 1 (row) · ■ Player 2 (column)

WORKED SOLUTION

- If P2 plays **L** → P1 best is **T** ($2 > 1 > 0$).
- If P2 plays **C** → P1 best is **B** ($2 > 1 > 0$).
- If P2 plays **R** → P1 best is **B** ($2 > 1 > 0$).
- Likewise, if P1 plays **B** → P2 best is **R** ($2 > 1 > 0$).

Only in cell **(B, R)** are both strategies best responses to each other → **unique Nash equilibrium (2, 2)**. No player can improve by unilaterally changing strategy.

Dominant strategies & the prisoner's dilemma

DEFINITIONS

- **Dominant strategy** — a strategy that is a player's best response no matter what the other player does. (It does *not* always exist!)
- **Dominant strategy equilibrium** — a Nash equilibrium in which *every* player is playing a dominant strategy.

Pest-control example → a prisoner's dilemma

Anil and Bala each choose how to deal with pest insects:

- **Toxic Tide (T)** — kills every insect but causes pollution (which spills onto the neighbour's farm).
- **Integrated Pest Control (IPC, "I")** — somewhat less effective, but no pollution.

		Bala	
		IPC	Toxic Tide
Anil	IPC	3, 3 social optimum	1, 4
	Toxic Tide	4, 1	2, 2 NASH (dominant)

■ Anil (row) · ■ Bala (column) · ■ equilibrium · ■ what both would prefer

WHY TOXIC TIDE IS DOMINANT

- If Bala plays **IPC**: Anil gets 4 (Toxic) vs 3 (IPC) → Toxic is better.
- If Bala plays **Toxic**: Anil gets 2 (Toxic) vs 1 (IPC) → Toxic is better.

Toxic Tide wins *regardless* of Bala → it is Anil's **dominant strategy** (and, by symmetry, Bala's). Both play their dominant strategy → equilibrium **(Toxic, Toxic) = (2, 2)**.

The twist: both would be *better off* at (IPC, IPC) = (3, 3). Individually rational behaviour makes them *collectively* worse off. That is the hallmark of a prisoner's dilemma.

DEFINITION — PRISONER'S DILEMMA

A game with a **dominant-strategy equilibrium** in which playing the dominant strategy yields **lower individual and total payoffs** than some other (cooperative) outcome. Originally framed with two burglars interrogated separately by police: each is tempted to confess, and both end up worse off than if they had stayed silent.

EXAM TRAP

A prisoner's-dilemma outcome *is* a Nash equilibrium (nobody can improve by deviating alone). It is *not* Pareto efficient, because both improve if they deviate together. "Nobody can improve alone" and "nobody can improve at all" are different statements — do not confuse them.

Real-world illustration: "Split or Steal" (TV show *Golden Balls*)

Two contestants divide a jackpot. Each secretly chooses **Split** or **Steal**:

- Both **Steal** → each gets **0**.
- One Steals, other Splits → the stealer takes **everything**, the splitter gets nothing.
- Both **Split** → each gets **half**.

"Steal" weakly dominates, so pure self-interest predicts (Steal, Steal) → both get nothing. Yet in practice (**Split, Split**) is **common** — a first hint that real people are not purely self-interested. (The famous clip where a player *talks* his opponent into cooperating shows how communication and social norms bend the outcome.)

Why did the theory predict the bad outcome? (roadmap to the cures)

The pessimistic prediction rests on three assumptions. Relax each one and you get one of the three "cures" that structure the rest of the unit:

Assumption that produces the dilemma	The corresponding cure
1. Players care only about their own payoffs .	→ Introduce social preferences / altruism (§D-I)
2. Nobody can make players pay for the consequences of their actions on others.	→ Introduce repeated games, social norms & peer punishment (§D-II)
3. Players cannot coordinate their actions beforehand.	→ Change the rules of the game — institutions & policy (§D-III)

Check-yourself question. Which statement is correct?

- An equilibrium in dominant strategies is always a Nash equilibrium.
- The outcome of a prisoner's dilemma is never a Nash equilibrium, because all players can improve their payoffs by changing their strategies.
- A combination of strategies that maximises the sum of all players' payoffs is always a Nash equilibrium.

Answer: (a). A dominant-strategy equilibrium is by construction a Nash equilibrium (each dominant strategy is a best response to everything). (b) is false — the PD outcome *is* a Nash equilibrium; players can only improve by deviating *together*, not unilaterally. (c) is false — the social optimum (e.g. IPC, IPC) maximises the total but is *not* a Nash equilibrium.

C. Evaluating outcomes — the Pareto criterion

We can predict what *will* happen (Nash). We also need a yardstick for whether an outcome is *good*. The standard economic yardstick is due to **Vilfredo Pareto**.

DEFINITIONS — PARETO

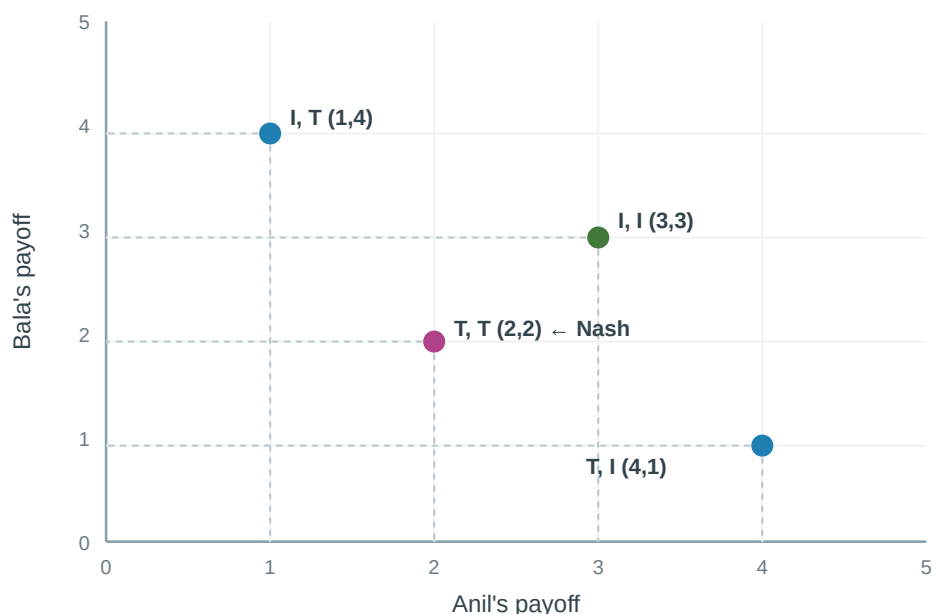
- **Pareto domination:** allocation A *Pareto-dominates* B if at least one party is better off under A than B and nobody is worse off.
- **Pareto efficiency:** an allocation is *Pareto efficient* if nobody can be made better off without making someone else worse off — there is no "free" improvement left.

INTUITION

Pareto efficiency = "no money left on the table." If you could make someone happier at no cost to anyone, the current outcome was wasteful. Note that **many** outcomes can be Pareto efficient at once: every way of splitting €1000 between two people is Pareto efficient, as long as none of it is thrown away.

Plotting the four pest-control outcomes

Put Anil's payoff on the horizontal axis and Bala's on the vertical axis. The four cells of the pest-control game become four points:



I, I = both use IPC; *T, T* = both use Toxic Tide; *I, T* = Anil IPC / Bala Toxic; *T, I* = Anil Toxic / Bala IPC. Notice *(I, I)* sits up-and-right of *(T, T)*: it Pareto-dominates the Nash outcome. Yet the players still land on *(T, T)*.

READING THE GRAPH

(I, I) Pareto-dominates (T, T) — it lies both further right (better for Anil) and higher (better for Bala). So the equilibrium the players actually reach is Pareto *inefficient*. The corners **(I, T)** and **(T, I)** are Pareto *efficient* (you can't move without hurting one of them), but they are extremely unequal.

Caveats — what Pareto efficiency does *not* tell you

- There is usually **more than one** Pareto-efficient allocation, and the criterion cannot rank them against each other.
- The criterion may not even compare an efficient with an inefficient outcome: e.g. **(T,I) is Pareto efficient but does not dominate (T,T)**, because moving from (T,T) to (T,I) makes Bala worse off.
- **Pareto efficiency is unrelated to fairness.** Keeping €99.99 of found €100 and giving your friend 1 cent is Pareto efficient — and obviously unfair.

D-I. Resolving dilemmas — social preferences & altruism

Recall cure #1: dilemmas arise *because players care only about their own payoffs*. But in experiments many people choose the "dominated" cooperative action anyway — they display **altruism**.

DEFINITION — ALTRUISM

Placing positive weight on *other people's* payoffs in your own preferences. If altruism is **sufficiently strong**, the socially desirable outcome can be reached even in a one-shot game — e.g. (Split, Split) in Golden Balls, or high contributions in public-good games.

The public-good game (a prisoner's dilemma for n players)

DEFINITION — PUBLIC GOOD

A good whose use by one person does **not** reduce its availability to others, so investment benefits everyone. Examples: national defence, street lighting, clean air. Because everyone benefits whether or not they paid, public goods invite **free riding**.

Set-up. Four farmers decide whether to contribute to an irrigation project. Contributing costs each farmer **\$10**; every \$10 contributed raises the crop yield on *each* of the four farms by **\$8**.

WHY NOBODY CONTRIBUTES (THE FREE-RIDER LOGIC)

If you contribute \$10, the total benefit created is $4 \times \$8 = \32 (great for society), but *your own* share is only \$8 — while you paid \$10. Privately you **lose \$2** per contribution, so a self-interested farmer contributes nothing. All four reason identically → nobody contributes → the \$32-per-contribution surplus is lost. Socially optimal: everyone contributes. Privately optimal: everyone free-rides. A classic n -player prisoner's dilemma.

The general payoff formula

Each player receives T tokens and chooses how many to keep (z_i) versus contribute to the common pool ($g_i = T - z_i$). Player i 's payoff is:

$$\text{payoff}_i = z_i + \frac{\beta}{n} (g_1 + g_2 + \dots + g_n)$$

Your payoff is what you kept, *plus* your equal share of the value the whole pool creates. The parameter β ("beta") sets how valuable a contributed token is:

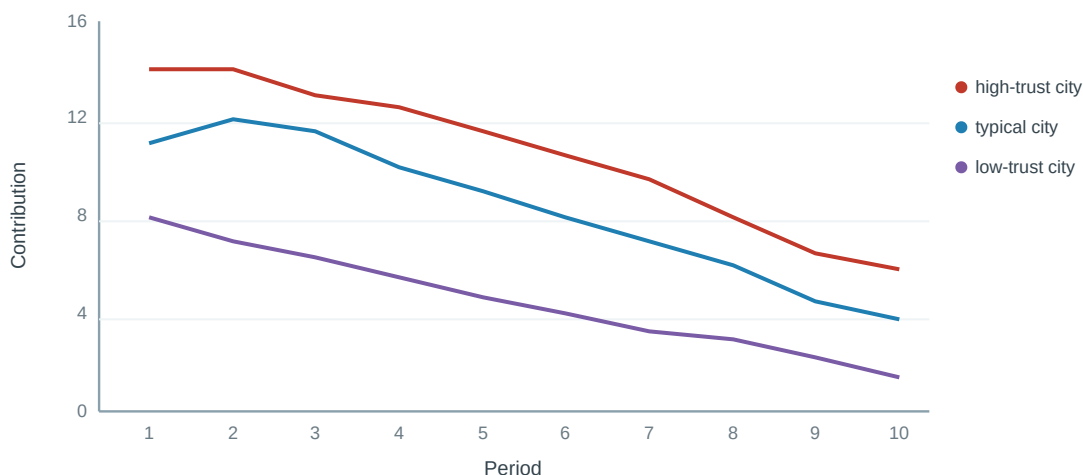
Condition	Consequence
$\beta/n < 1$ (i.e. $1 > \beta/n$)	A purely self-interested player keeps all tokens — each token contributed returns less than \$1 to you.
$\beta > 1$	It is socially optimal to invest all tokens — each token creates more than \$1 of total value.
$1 < \beta < n$	Both hold at once → private and social incentives conflict → the game is a social / prisoner's dilemma .

What experiments actually find

- Contributions are **substantial but not universal** — on average about **50%** of tokens in round 1; some players keep almost everything.
- When the game is **repeated**, contributions **decay** — from ~50% in round 1 down toward ~10% by round 5, as cooperators get frustrated by free-riders.
- Lower $\beta \rightarrow$ lower contributions.
- Economics / business students appear to contribute *less* — either they are *taught* to be more self-interested, or more self-interested people *select into* economics. (An open, much-debated question.)

Public-good game: contributions fall as the game repeats

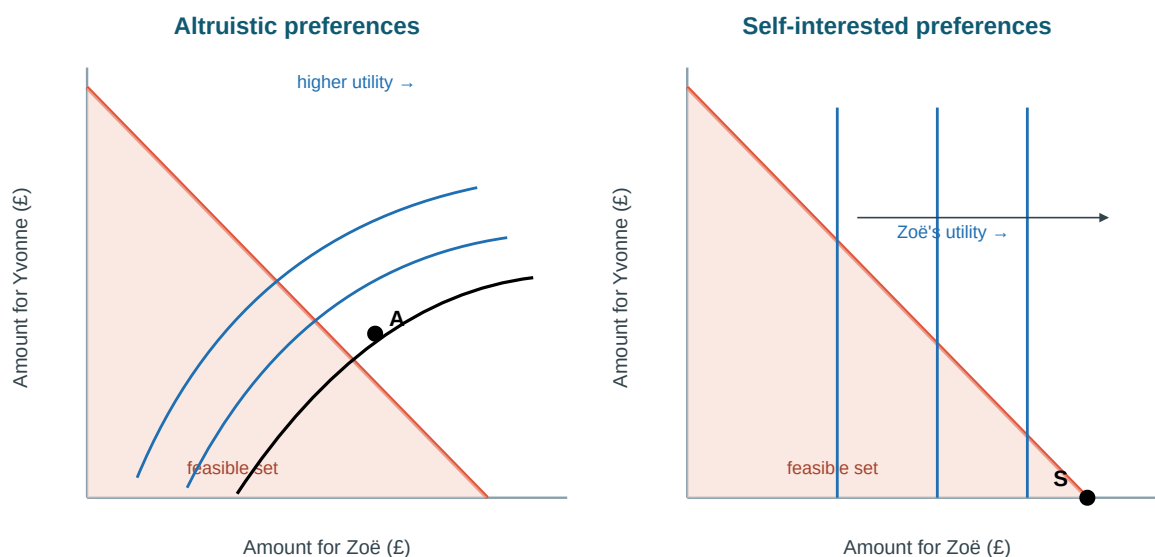
(20 tokens each, $\beta = 1.6$, $n = 4$ — representative of the multi-country experiment)



Every group starts fairly cooperative, but contributions erode round after round — the decay that motivates the "peer punishment" fix in *SD-II*. (Schematic of Herrmann–Thöni–Gächter's 16-city experiment.)

Modelling altruistic preferences with indifference curves

How do we *represent* altruism formally? Reuse the earlier-unit tool: **indifference curves + a feasible set**. Zoë wins £200 in a lottery and decides how much to keep vs. share with her flatmate **Yvonne**. The budget line (*feasible frontier*) runs from (£0 for Zoë, £200 for Yvonne) to (£200, £0): every pound Zoë keeps is a pound Yvonne loses.



Left — altruistic Zoë: she cares about Yvonne too, so her indifference curves are convex and downward-sloping. The best feasible point **A** is an interior tangency — she shares some money. **Right — self-interested Zoë:** she cares only about her own amount, so indifference "curves" are vertical lines (more for Zoë is always better, Yvonne's amount is irrelevant). The optimum **S** is the corner — she keeps everything.

THE PUNCH-LINE

Altruism changes the *shape* of the indifference curves, which moves the optimum from the selfish corner (S) to an interior sharing point (A). Formally, altruistic players are maximising a different utility function — one that puts weight on others' payoffs — so the very same feasible set delivers a more cooperative choice.

Altruism inside the pest-control game

The same idea rescues the pest-control dilemma. If Anil is **self-interested**, his indifference "curves" over the four outcomes are vertical lines (only his own income matters) and he still ends at the Toxic-Tide equilibrium. If Anil is **altruistic** — his indifference curves become downward-sloping because he now dislikes polluting Bala — then IPC can become his best response and the cooperative outcome (I, I) can emerge as an equilibrium. Preferences, not just payoffs, determine the outcome.

D-II. Resolving dilemmas — repeated games & punishment

Cure #2: in a **one-shot** game nobody can make you pay for harming them. But most real interactions **repeat**. When they do, social norms, **reciprocity** and **peer punishment** become possible.

KEY IDEA — REPEATED GAMES

Behaving selfishly in one period has **consequences in future periods** (others can retaliate). So the selfish action **may no longer be a dominant strategy** once the future is taken into account. Cooperation can be sustained as an equilibrium of the repeated game even though it is not an equilibrium of the one-shot game.

Example: two petrol stations (a repeated cartel dilemma)

Stations **Jupiter** and **Saturn** each choose to **Maintain** a high price or **Cut** it. Weekly profits:

		Saturn	
		Maintain	Cut
Jupiter	Maintain	1000 / 1000 <i>both prefer</i>	700 / 1300
	Cut	1300 / 700	800 / 800 <i>one-shot Nash</i>

■ Jupiter (row) · ■ Saturn (column)

One-shot: Cutting is a dominant strategy for both → outcome (Cut, Cut) = (800, 800), even though both prefer (Maintain, Maintain) = (1000, 1000). This is the *cartel's dilemma*.

THE TRIGGER STRATEGY (HOW REPETITION FIXES IT)

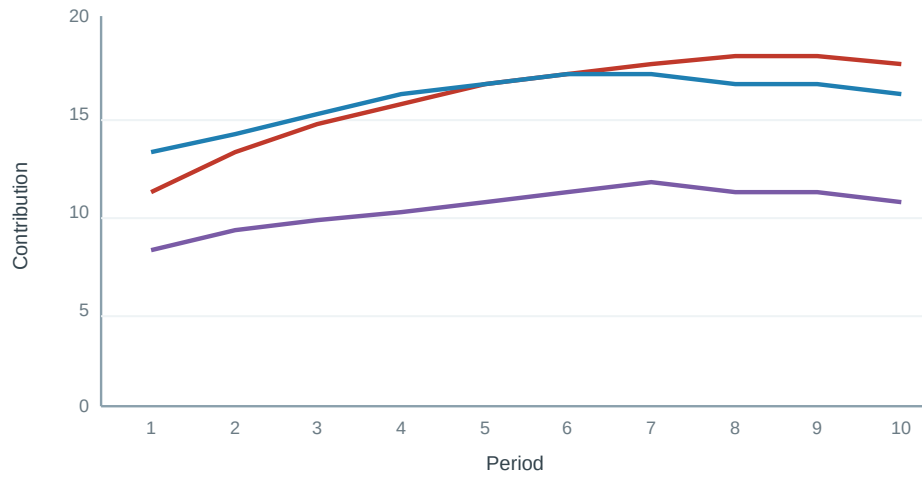
Play a "grim/trigger" strategy: *start with the high price, and keep it as long as the rival keeps a high price. If the rival ever cuts, retaliate by cutting for a stretch of periods before returning to the high price.*

This creates a **trade-off**: cutting earns a one-time bonus *today* (1300 vs 1000), but triggers punishment that lowers profit *tomorrow*. Cooperation is an equilibrium **if players value future profits highly enough** — i.e. the future is not discounted too steeply — because only then does the threat of punishment outweigh the short-term temptation.

Peer punishment in the public-good game

Return to the public-good experiment. When a **punishment option is added** — players can pay a small cost to fine free-riders — contributions no longer collapse. Instead they **rise and stay high** across periods, the mirror image of the decay we saw without punishment.

Add peer punishment → contributions climb instead of collapsing



Compare with the earlier decaying graph: the only change is that cooperators can now punish free-riders. The threat of punishment sustains high contributions — the same mechanism as the petrol-station trigger strategy.

D-III. Resolving dilemmas — changing the rules (sequential games)

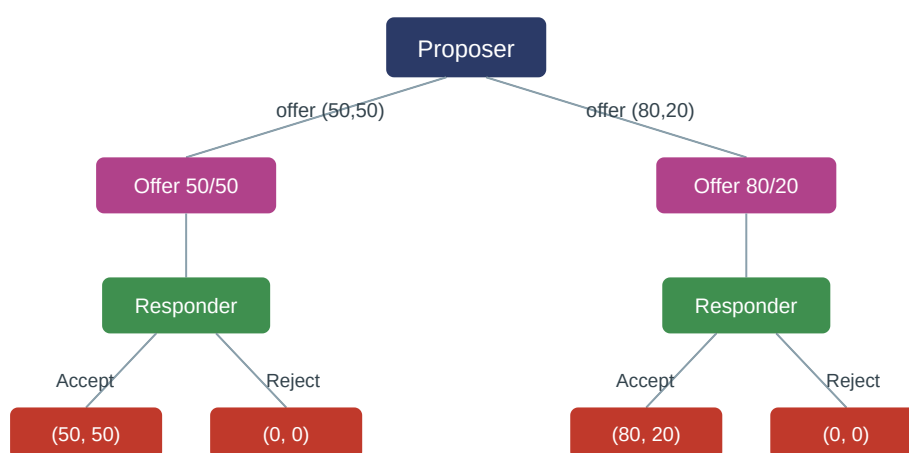
Cure #3: alter the **rules of the game** itself — the order of play, who can commit, what is enforceable. A leading example of how rules matter is the **ultimatum game**, our first **sequential** (rather than simultaneous) game.

DEFINITION — THE ULTIMATUM GAME

A **sequential** game dividing an **economic rent** (e.g. a cash prize). The **Proposer** offers the **Responder** a share; the Responder *observes* the offer and then **accepts** (both get the proposed split) or **rejects** (both get nothing).

The game tree

Sequential games are drawn as **trees**: nodes are decision points, branches are actions, and the numbers at the bottom are the (Proposer, Responder) payoffs.



Payoffs shown as (Proposer, Responder). Reject → both get 0.

The Proposer moves first and can foresee the Responder's reaction. A purely self-interested Responder accepts any positive amount ($20 > 0$), so the self-interested prediction is a tiny offer that is accepted.

THEORY VS. BEHAVIOUR

Self-interested prediction: the Proposer offers (almost) nothing and the Responder accepts. **What people actually do:** Proposers usually offer close to a **50–50 split**, and Responders **reject** low offers — paying a personal cost to punish unfairness. This reveals **fairness, inequality aversion, social norms and reciprocity** operating on top of raw payoffs.

Evidence: Kenyan farmers vs. US students

When the ultimatum game is run in different populations, offers cluster well above zero and are consistent with **social preferences** — though also, importantly, with **expected-payoff maximisation** once you account for the fact that low offers get rejected. For example, Kenyan-farmer responders were expected to reject a 30% offer about **48%** of the time but a 40% offer only **4%** of the time — so offering 40% can be smart *even for a purely selfish proposer*, because it avoids rejection. Roughly half of Kenyan-farmer proposers offered 40%.

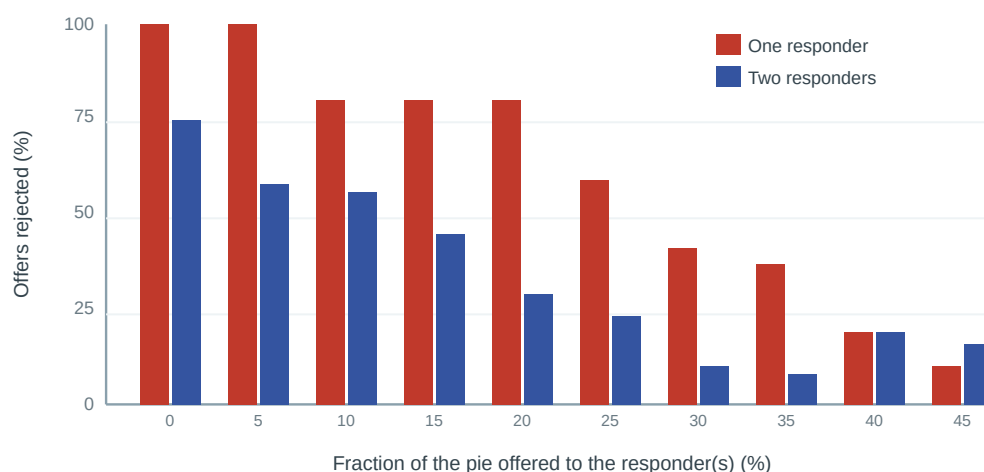
SUBTLE BUT EXAM-WORTHY

Generous offers are consistent with *two* different stories: (1) proposers genuinely care about fairness, or (2) proposers are selfish but rationally avoid rejection. Behaviour alone often cannot distinguish them — a recurring theme in behavioural economics.

The rules of the game matter

Change the rules and the outcome shifts predictably. Put **two responders in competition** (only one can accept, so if one rejects the other may still take the deal) and rejection of low offers **falls sharply**. Competition among responders pushes the outcome **back toward the self-interested prediction**: proposers can get away with meaner offers.

Competition between responders → fewer rejections



At every offer level, adding a second, competing responder (blue) lowers the rejection rate versus a single responder (red). The rules — here, whether responders compete — reshape how much fairness the proposer must show.

POLICY READING OF "CHANGING THE RULES"

Governments and institutions resolve social dilemmas precisely by changing rules: taxes/subsidies that make people "pay" for external effects, binding treaties, property rights, regulation. Each rewrites the payoffs or the order of play so that individually rational behaviour coincides with the social optimum.

E. Coordination games

So far the problem was that the equilibrium was *bad*. A different problem arises when there are **several equilibria** and players must somehow land on the same one.

COORDINATION PROBLEM

When a game has **more than one Nash equilibrium** and players choose independently, the **socially optimal equilibrium may not be selected**. Society can get "**stuck**" in a worse equilibrium — because, being an equilibrium, no one gains by deviating *alone*.

A coordination game between the two farmers

Now Anil and Bala benefit from **specialising in different crops** (so they can trade). There are two Nash equilibria — but one is better for both.

		Bala	
		Rice	Cassava
Anil	Rice	0, 1 Nash — but worse	2, 2 Nash — but worse
	Cassava	4, 4 Nash — better	1, 0

■ Anil (row) · ■ Bala (column) · ■ sub-optimal Nash · ■ optimal Nash

TWO EQUILIBRIA — VERIFY

Both **(Rice, Cassava) = (2,2)** and **(Cassava, Rice) = (4,4)** are Nash: in each, given the other's crop, neither farmer wants to switch. They coordinate on *different* crops in both. But (4,4) Pareto-dominates (2,2). If they happen to be stuck at (2,2), **no unilateral move helps** — Anil switching alone would drop him from 2 to 1. They need to move *together*, i.e. to **coordinate**.

Example: climate change as a coordination game

China and the US each choose **Restrict** emissions or **Business-As-Usual (BAU)**. Ranking outcomes BEST > GOOD > BAD > WORST:

		US	
		Restrict	BAU
China	Restrict	BEST, BEST social optimum	GOOD, WORST
	BAU	GOOD, WORST	BAD, BAD other Nash

■ China (row) · ■ US (column)

- For purely self-interested players, **BAU is the dominant-strategy** instinct, and an **emissions treaty** would deliver the socially optimal (Restrict, Restrict).
- With **inequality aversion and reciprocity** (each will restrict only if the other does), the game has **two Nash equilibria**: (Restrict, Restrict) = BEST and (BAU, BAU) = BAD. The social optimum is *one* of them — so coordination (a treaty, mutual commitment) is what selects it.

Driving on the left or the right

The purest coordination game. Each driver picks a side; matching is fine, mismatching is a crash.

		Driver 2	
		Left	Right
Driver 1	Left	0, 0 Nash	-10, -10
	Right	-10, -10	0, 0 Nash

■ Driver 1 (row) · ■ Driver 2 (column)

TWO EQUILIBRIA, AND WHY THE RULE IS DECISIVE

(Left, Left) and **(Right, Right)** are both Nash — neither is intrinsically better, but **coordination is essential**. Which one a country lands on is a matter of convention and history. Most of the world drives on the right; a minority (UK, India, Australia, Japan, much of southern Africa...) on the left. When Sweden switched from left to right overnight on **"Dagen H" (3 September 1967)**, a central authority moved everyone from one equilibrium to the other simultaneously — a vivid demonstration that escaping/selecting an equilibrium can require a coordinating institution.

(The **Dvorak vs. QWERTY keyboard** is another classic: society can stay locked into an inferior standard simply because everyone has coordinated on it.)

Summary

1. **Social interactions can be modelled as games.** Players choose **best responses** to others' strategies; a **Nash equilibrium** is a mutual-best-response profile where no one gains by deviating alone.
2. **Social dilemmas** (e.g. the prisoner's dilemma) have equilibria that are *not* socially optimal. They can be resolved by: **social preferences / altruism**, **repeated interaction with peer punishment**, or **binding agreements / changing the rules**. The **rules of the game** shape outcomes.
3. **Multiple Nash equilibria** create **coordination problems**; society can get stuck in a bad equilibrium. **Economic and political institutions** help select the socially optimal outcome.

Glossary — quick revision

Term	One-line meaning
Social dilemma	Independent self-interest → socially sub-optimal outcome.
Tragedy of the commons	Shared resources get over-exploited.
Free riding	Enjoying others' contributions without contributing.
Strategy / dominant strategy	An action / an action that is best <i>whatever</i> others do.
Best response	Highest-payoff action <i>given</i> the others' choices.
Nash equilibrium	Everyone best-responding at once; no unilateral gain.
Dominant-strategy equilibrium	A Nash equilibrium of dominant strategies.
Prisoner's dilemma	Dominant-strategy equilibrium that is worse for all than cooperating.
Pareto domination	A makes someone better off and no one worse off than B.
Pareto efficiency	No one can be made better off without hurting someone.
Altruism / social preferences	Caring about others' payoffs, not just your own.
Public good	Non-rival good; one person's use doesn't reduce others' availability.
Repeated game / trigger strategy	Interaction repeats; retaliation can sustain cooperation.
Ultimatum game	Sequential split-the-rent game; proposer offers, responder accepts/rejects.
Coordination game	Multiple equilibria; players must land on the same one.

Self-test questions

1. In the pest-control game, explain why (Toxic, Toxic) is a Nash equilibrium even though both players prefer (IPC, IPC).

Because from (Toxic, Toxic) neither can improve by switching *alone* (Toxic gives 2 vs 1 if you unilaterally move to IPC). The improvement to (IPC, IPC) requires *both* to move together, which the equilibrium concept does not allow.

2. A public-good game has $T = 20$ tokens per player, $n = 4$, $\beta = 1.6$. Is contributing individually rational? Is it socially optimal?

Each contributed token returns $\beta/n = 1.6/4 = 0.4$ to *you* — less than 1 — so a selfish player keeps everything (not individually rational to contribute). But each token creates $\beta = 1.6 > 1$ of *total* value, so full contribution is socially optimal. Private and social incentives conflict → prisoner's dilemma.

3. Why can a low ultimatum-game offer be rejected even by a "rational" responder — and how does adding a second responder change this?

A responder with *social preferences* (fairness / inequality aversion) gets negative utility from an unfair split and will pay the cost of rejecting to punish it. Adding a competing second responder means rejecting no longer denies the proposer the deal (the other can accept), so rejection becomes costly and pointless — rejection rates fall and offers drift toward the self-interested outcome. The rules of the game changed the behaviour.

4. Driving-side and climate-treaty games both have two Nash equilibria. What is the key difference?

In driving, the two equilibria are equally good — the only thing that matters is *coordinating on the same one*. In the climate game the two equilibria differ in welfare (BEST,BEST vs BAD,BAD), so coordination must not only match players but steer them to the *better* equilibrium — hence the role of a treaty/institution.